



Reservoir Spiking Neural Networks for Accurate State-of-Charge Estimation in Battery Management Systems

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Cite this study: Kamarudin, MR., Zainudin, MNS., Mispan, MS & Sofian, H. (2026). Reservoir Spiking Neural Networks for Accurate State-of-Charge Estimation in Battery Management Systems. Turkish Journal of Engineering, 10 (2), 407-417.

<https://doi.org/10.31127/tuje.1804828>

Keywords

Reservoir spiking neural network
State-of-charge prediction
Neuromorphic computing
Battery management system

Abstract

The growing integration of electric vehicles and renewable energy systems has heightened the need for efficient and reliable battery management systems (BMS). Accurate state-of-charge (SoC) estimation is a crucial element in maintaining battery performance and longevity, yet conventional model-based and neural network approaches often struggle to capture complex and nonlinear battery dynamics. To address these limitations, this study presents a proof of concept based on a Reservoir Spiking Neural Network (RSNN) designed to improve SoC estimation accuracy through biologically inspired computation. The proposed model combines the temporal processing strength of spiking neural networks with the dynamic memory of reservoir computing, enabling the network to effectively represent temporal dependencies in battery discharge patterns. Different spike encoding strategies were explored to optimise the transformation of continuous battery data into spiking activity. The analysis reveals that population-based encoding enhances prediction accuracy by enriching the internal spiking representation within the reservoir, though it requires slightly higher synaptic activity. Experimental results demonstrate that the RSNN provides competitive SoC estimation performance during critical discharge stages, without a significant increase in training time. These findings establish RSNNs as a promising alternative for SoC estimation, offering improved adaptability, high computational efficiency, and suitability for low-power embedded implementations in advanced BMS applications.

Research Article

Received: October 16, 2025

Revised: January 4, 2026

Accepted: March 2, 2026

Published:30.04.2026



1. Introduction

The rapid rise of electric vehicles and renewable energy systems has intensified the demand for efficient and reliable battery management systems (BMS) [1]. This growing interest extends across multiple areas, such as advancements in wireless power transfer technologies for Electrical Vehicle (EV) charging stations [2]. Among the critical parameters monitored by BMS, the State of Charge (SoC) represents the remaining energy as a percentage of full capacity, making it indispensable for performance optimization and battery lifespan extension

[3]. Traditional SoC estimation methods, such as Open Circuit Voltage (OCV) and Coulomb counting, face several practical limitations [4–7]. OCV requires the battery to be at rest, preventing real-time updates, while Coulomb counting accumulates measurement errors over time and requires periodic recalibration. These shortcomings restrict their applicability in dynamic, real-time environments. Advanced techniques such as Kalman Filters [8–11], optimization-based methods [12,13], and deep learning models [14–18] have improved estimation accuracy but often demand significant computational resources, limiting their suitability for resource-

constrained, edge-based BMS platforms. Similar computational efficiency challenges have been addressed in other intelligent embedded systems, such as the design of cost-effective robotic vacuum cleaners [19] and the application of unmanned aerial vehicles (UAV) for cultural heritage documentation [20] and agricultural monitoring [21]. These studies emphasize the growing importance of energy-efficient and real-time computation in autonomous systems, underscoring the need for lightweight and adaptive neural models in energy management applications.

In this context, Spiking Neural Networks (SNN) have emerged as a promising alternative to conventional deep learning architectures. As the third generation of neural networks, SNN process information through discrete spikes, emulating the behavior of biological neurons [22,23]. Unlike traditional artificial neural networks (ANN) that rely on continuous activations, SNN neurons fire only when stimulated, making them inherently event-driven and energy-efficient. This property allows for reduced computational demands and lower power consumption which is an advantage for real-time embedded systems. Moreover, their ability to capture temporal dynamics from event-based data enables efficient spatio-temporal feature extraction, a crucial aspect for accurate battery state monitoring. In battery research, however, applications of SNN remain limited, with only preliminary studies demonstrating their potential for battery state estimation, including convolutional SNN-based approaches that highlight energy-efficiency benefits but do not exploit reservoir-based spiking dynamics [24]. However, existing SNN implementations face challenges such as reduced prediction accuracy, primarily due to lossy encoding of continuous input data into spike trains and the limited maturity of training algorithms for spiking dynamics [25].

A promising approach to address these limitations is the Reservoir Spiking Neural Network (RSNN), which integrates the principles of Reservoir Computing (RC) with spiking neuron architectures for efficient temporal signal processing. In this framework, a fixed recurrent reservoir projects input data into a high-dimensional temporal space, while only the readout layer undergoes training. This structure significantly reduces computational complexity and enhances the model's suitability for low-power, real-time applications such as SoC estimation. To the best of the authors' knowledge, a systematic investigation of RSNN for battery SoC estimation, particularly focusing on the impact of spike encoding strategies on both prediction accuracy and energy-related behavior, has not yet been reported.

Motivated by these advantages, this study proposes an RSNN-based approach for SoC estimation that combines the strengths of both SNN and RC frameworks to achieve accurate yet computationally efficient performance. Specifically, this work provides a detailed analysis of how different spike encoding techniques influence reservoir dynamics, estimation accuracy, and energy-related characteristics. The main contribution of this work lies in developing a reservoir-based spiking neural model for battery SoC estimation, emphasizing the effects of encoding strategies and activation functions

on prediction accuracy and computational efficiency. The remainder of this paper is organized as follows: Section 2 details the proposed methodology, Section 3 presents the results, followed by a comprehensive discussion in Section 4. Finally, Section 5 concludes the study and outlines potential directions for future research.

2. Method

2.1. Battery dataset

In this study, we utilize a dataset provided by the National Aeronautics and Space Administration (NASA) [26]. The dataset contains cycle aging data for pouch-type LCO/graphite 1.5 Ah cells, including measurements of current, voltage, charge/discharge capacity and energy, internal resistance, and impedance. The cells were aged through cycling at low C-rates, followed by calendar aging under various storage temperatures. The dataset is publicly available on Kaggle and comprises experimental data from four lithium-ion batteries (#5, #6, #7, and #18).

2.2. Data preprocessing

The data preprocessing procedure followed a structured pipeline to prepare the NASA battery dataset for RSNN model training. Key battery parameters [*capacity*, *voltage_measured*, *current_measured*, *temperature_measured*, *current_load*, *voltage_load*, *time*] were extracted from the discharge cycles, which were prioritized due to their relevance in accurate SoC computation. These features were selected to capture complementary electrical and thermal behaviors of the battery, allowing the RSNN to learn both instantaneous responses and long-term discharge dynamics without imposing prior feature bias. To ensure uniform feature scaling and prevent dominance by any single feature, all variables were normalized using the Min-Max scaling technique. The normalized input features are then encoded as spike streams to fed to the input layer. For this work, different encoding techniques are used as follows:

- **Fixed Rate Encoding:** The input values are converted into deterministic spike trains by mapping each input intensity to a fixed firing frequency. Values within the normalized range $[0, 1]$ are linearly scaled by a predefined maximum firing rate (*max_rate*) to determine the corresponding spike rate. For example, an input of 0.5 with a *max_rate* of 100 Hz produces a firing rate of 50 Hz. This rate is then expressed as a fixed inter-spike interval, ensuring that the neuron generates spikes at uniform, evenly spaced time steps throughout the simulation window.
- **Poisson Encoding:** This approach mimics the stochastic firing behavior of biological neurons by converting normalized continuous inputs into randomly occurring spike trains. Each input value, scaled between $[0, 1]$, is multiplied by the maximum firing rate to estimate the expected number of spikes within the simulation period. For instance, an input of 0.7 produces a higher expected firing rate than 0.2, resulting in more frequent spiking activity. Unlike

fixed-rate encoding, spike timings are governed by probabilistic dynamics: the average inter-spike interval (*avg_ISI*) determines the mean temporal spacing, while variability in spike occurrence creates irregularly distributed events across time.

- **Population Encoding:** The input value is represented as a distributed activation pattern across a group of neurons (specified by *neurons_per_input*). A single normalized input (e.g. 0.3) is encoded using a population of N neurons, each with a preferred stimulus value, or center. The response of each neuron is computed using a Gaussian tuning curve centered on its preferred value, producing higher activations for neurons whose centers are closer to the input. For example, with an input of 0.3, neurons centered around 0.29, 0.36, and 0.21 exhibit stronger activations, resulting in a smooth localized response across the population. This representation increases input resolution, introduces redundancy for robustness against noise, and enables the reservoir to capture fine-grained variations in the signal through diverse spatiotemporal activation patterns.

The selected encoding strategies represent different spike-generation paradigms, enabling a systematic evaluation of how spike diversity and firing statistics influence prediction accuracy and energy efficiency in RSNN-based SoC estimation.

2.3. Architecture of reservoir spiking neural network (RSNN)

The RSNN consisted of three layers: input layer, reservoir network and readout network as shown in Figure 1. The input layer receives encoded features from the battery dataset. Its size depends on the number of features and the encoding method. In population encoding, each feature is mapped to a group of 10 neurons, giving $N \times 10$ input neurons. In Poisson and fixed-rate encoding, a one-to-one mapping is used, requiring only N neurons. Here, $N = 7$ input neurons are used, with each neuron corresponding to one feature. Each continuous feature is converted into a spike train tensor of shape $[T, N]$, where T is the simulation time (e.g., 100 ms). These spike trains are projected to the reservoir through sparse synaptic connections, with connectivity defined by a density parameter.

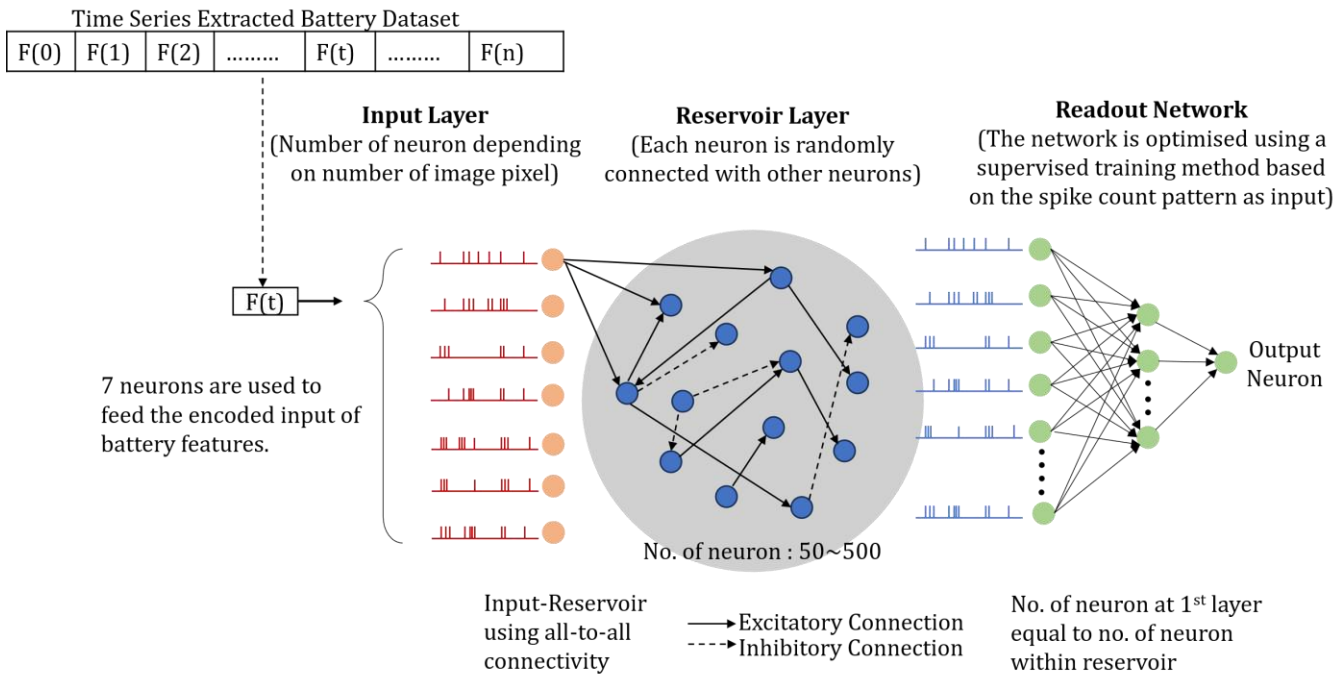


Figure 1. Reservoir spiking neural network (RSNN) architecture. the network model consists of three main parts; the input layer, reservoir layer and readout network

Synaptic weights are initialized from a statistical distribution to promote diverse dynamics. Connected to the input layer is the reservoir network. This network is the computational core of the RSNN, composed of n_nodes of Leaky Integrate-and-Fire (LIF) neurons. Here, the model was tested with different values of n_nodes in the range of 50–500 neurons. It nonlinearly projects input spike trains into a high-dimensional space, providing short-term memory for temporal tasks such as battery SoC estimation. LIF neurons integrate postsynaptic potentials and fire when their membrane potential exceeds a threshold. In the absence of input, the potential decays exponentially toward rest. This leak

introduces fading memory, while the reservoir's sparse, randomized connectivity ensures rich internal dynamics.

The pattern captured by reservoir network later is stored and trained with supervised learning technique on the Readout network. The readout network decodes reservoir activity into a scalar SoC output. Reservoir spikes over the simulation window are summed to form a feature vector, with dimension equal to the number of reservoir neurons (e.g., $n_nodes = 100$). This vector captures firing profiles in a static form suitable for regression. A multilayer feedforward neural network serves as the regression model, trained only in the readout layer while reservoir connections remain fixed.

Training is performed over n_epochs using Mean Squared Error (MSE) loss and Stochastic Gradient Descent (SGD). The parameter ranges were chosen based on commonly adopted configurations in reservoir computing and spiking neural network literature, while remaining within computationally feasible limits for embedded and neuromorphic deployment.

3. Results

3.1. Analysis of input and reservoir spike patterns

Different encoding techniques generate distinct spike patterns at the input layer. These patterns directly influence the spiking behavior within the reservoir layer. In the RSNN model, training and optimisation are based on the spike count patterns of reservoir neurons. For the network to be optimised effectively, the spike counts across reservoir nodes must be distinguishable for different input values. To investigate this, the spike patterns in both the input and reservoir layers were

analysed using four encoding approaches: Poisson, Fixed, Population-Poisson, and Population-Fixed. The analysis was carried out with input (I) values of 0.25, 0.5, 0.75, and 1.0. As described in the previous section, population encoding represents each input feature with multiple N nodes. Each node holds a specific value that is then transformed into a spike train using either Poisson or Fixed encoding.

Figure 2 depicts the spike patterns at the input layer for different input values. Figure 2(a) and Figure 2(b) show the spike patterns when the input is encoded using Poisson and Fixed encoding techniques. In the Fixed encoding technique, the spike interval is determined by the maximum spike rate, which is set to half of the timestep length. As a result, both the number of spikes and the interval are linearly proportional to the original input value. In contrast, for Poisson encoding, the spike interval is governed by the avg_ISI parameter. This parameter is configured such that, at maximum input value, the generated spike train is broadly distributed across the entire timestep window.

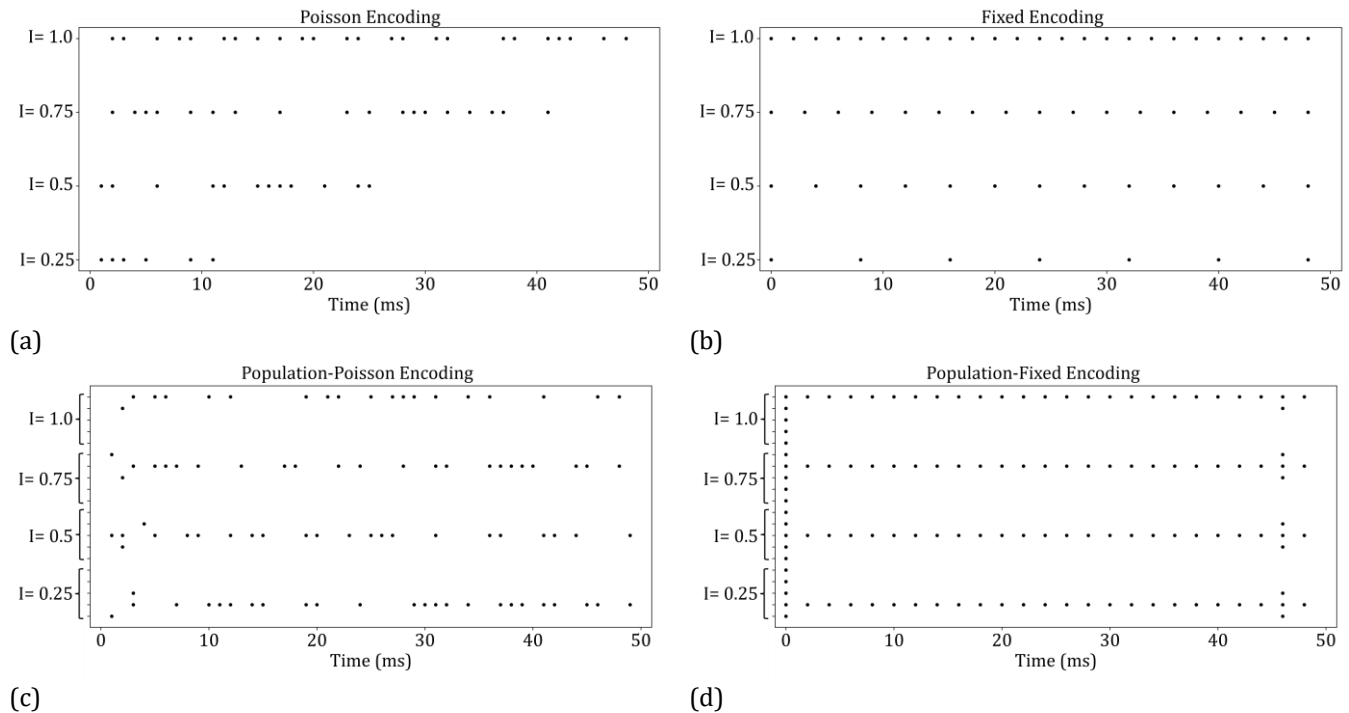


Figure 2. Spike pattern on input layer with different encoding techniques. (a) poisson (b) fixed. (c) population-poisson (d) population-fixed

In Figure 2(c) and (d), each input feature is represented using multiple N nodes through population encoding. With this approach, spikes are distributed across the entire timestep window, while the spike timing and the nodes in which spikes occur depend on the input value. For example, with population encoding at $N=5$, when the input value is $I=1.0$, $node_5$ generates a dense spike train. In contrast, when the input is $I=0.25$, $node_1$ produces a denser spike train compared to the others. This mechanism introduces greater diversity in the spiking behavior of the reservoir layer, which is directly influenced by the input features.

This diversity is examined by measuring the spiking activity of the reservoir network, quantified through the total spike count of each node in the reservoir. Figure 3

presents the heatmap of spike counts across 50 nodes ($N=50$) for different encoding techniques. Figure 3(a) corresponds to the low input value of 0.25, while Figure 3(b) corresponds to the maximum input value of 1.0. At low input (0.25), both Poisson and Fixed encoding techniques fail to generate sufficiently diverse spike count patterns. This is confirmed by the low entropy value, poisson encoding at $H=0.21$ and fixed encoding at 0.09.

In contrast, population encoding improves diversity, with entropy values increasing above 0.4, and the Population-Fixed encoding showing the highest diversity in spike count patterns. A similar trend is observed at the maximum input value (1.0), where population encoding continues to produce high diversity in spike count

patterns, reflected in consistently higher entropy values. However, with Population-Fixed encoding, the system consistently exhibits high energy consumption, ranging approximately from 29.3 μJ to 32.2 μJ per input (assuming 10 nJ/spike event), as shown in Figure 4. This is evaluated by measuring the total number of synaptic events within a single timestep window for processing one input value. A higher number of synaptic events directly reflects higher power consumption in a

neuromorphic system. Switching to Population-Poisson encoding significantly reduces energy usage, ranging roughly from 20.5 μJ to 23.4 μJ , while providing a more efficient trade-off between spike diversity and power consumption. Since higher diversity is strongly associated with prediction accuracy, this trade-off is important: in the RSNN, the fully connected layer relies on spike count patterns for effective network optimization.

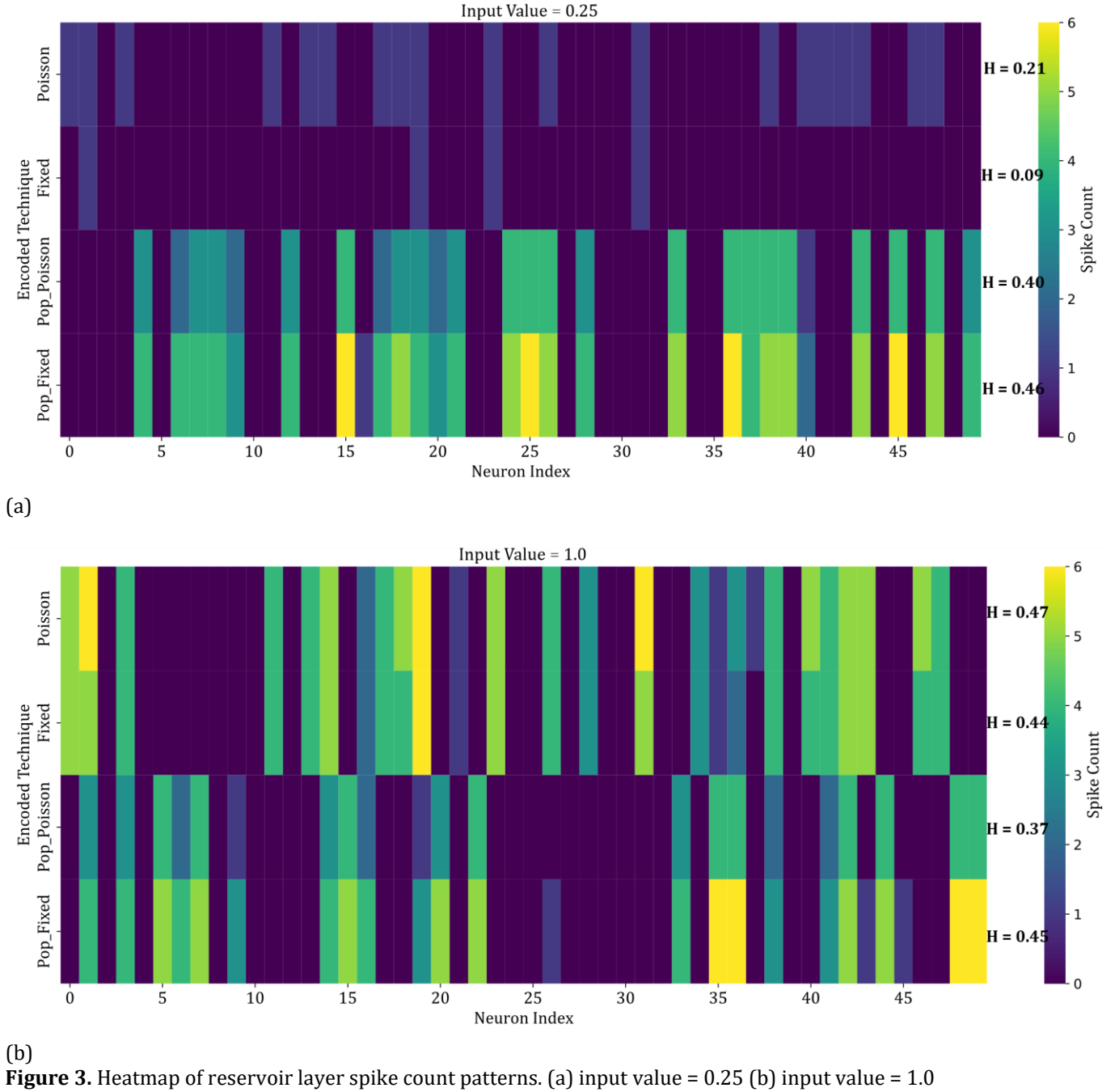


Figure 3. Heatmap of reservoir layer spike count patterns. (a) input value = 0.25 (b) input value = 1.0

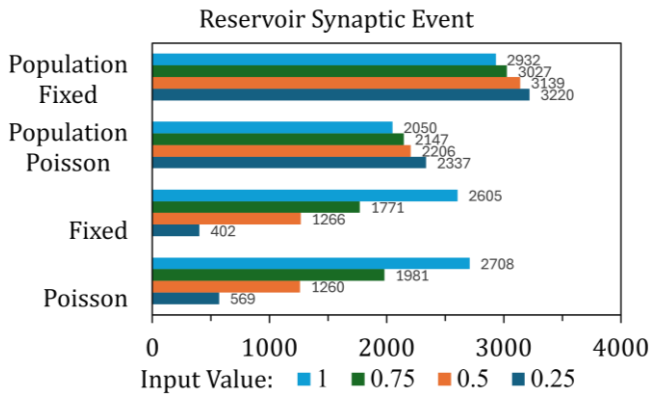


Figure 4. Reservoir synaptic events across different encoding techniques

3.2. RSNN Performance under Different Encoding Techniques

In this section, the accuracy of the RSNN is evaluated under different encoding techniques and network configurations. For the network configurations, the focus is placed on the number of reservoir nodes, the length of the timestep window and the activation function of the fully connected layer, as these are critical for model development on resource-constrained embedded platforms. Other network configurations are kept fixed across all simulations, as summarised in Table 1. Firstly, the accuracy of the RSNN is evaluated by varying the number of reservoir nodes. Table 2 summarises the performance accuracy of the RSNN for each type of encoding technique. As shown in the table, with population encoding, whether using Poisson or Fixed encoding, the network model is optimised more effectively, achieving lower average loss.

Table 1. Common network configuration setup

Item	Configuration
Input-Reservoir Connection	all-to-all
Input-Reservoir Weight	random with partial inhibitory weight
Reservoir-Reservoir Connection	partial connection
Input-Reservoir Weight	random with partial inhibitory weight
Epoch	20
Hidden Layer	1 with 10 nodes
Activation Function	Sigmoid
Time Step Window	50 ms

The prediction accuracy increases by approximately 26% with Population-Poisson and 39% with Population-Fixed, compared to their respective non-population counterparts. Additionally, it is evident that increasing the number of reservoir neurons does not provide any significant improvement, as 100 reservoir nodes are sufficient for the network configuration listed in Table 1. Next, the best network configuration is examined with different lengths of the time step window.

In spiking neural network models, the step window is critical as it directly affects the total training time of the network and the system's response time in real hardware. Table 3 shows the effect on the accuracy of the RSNN model using the Population-Fixed encoding technique. Theoretically, increasing the time step window increases the total spike count in the reservoir network, which enhances the diversity of the spike count pattern. As shown in the table, increasing the time up to 200 ms did not significantly improve accuracy. Similarly, reducing the time did not affect the accuracy much (only reduced by about 9%), although it sped up the training process by approximately 21%, as well as reducing the system response time to only 12 ms per input iteration on a conventional computer.

Table 2. RSNN accuracy with different encoding techniques and reservoir sizes

Neuron	50		100		250		500	
	Loss	RMSE	Loss	RMSE	Loss	RMSE	Loss	RMSE
Poisson	0.0027	0.062	0.0026	0.061	0.0031	0.057	0.0041	0.064
Fixed	0.0025	0.065	0.0027	0.071	0.0020	0.058	0.0019	0.057
Population Poisson	0.0013	0.049	0.0010	0.045	0.0010	0.046	0.0011	0.051
Population Fixed	0.0014	0.056	0.0010	0.043	0.0012	0.046	0.0838	0.187

Table 3. RSNN performance with different time step

Time Step[ms]	Training Time[s]	Loss	Latency [ms]	RMSE
30	233	0.0012	12	0.047
50	294	0.0010	14	0.043
100	460	0.0010	26	0.045
200	771	0.0022	48	0.053

Figure 5 depicts the comparison of SoC prediction using different encoding techniques with the RSNN model at the 10th and 150th battery cycles. Theoretically, the calculated SoC decreases linearly over time as the data is measured in a controlled laboratory environment. Since the RSNN relies on spiking activity to

convey information, the prediction generally follows the decreasing pattern but exhibits fluctuations. These fluctuations are more pronounced when the RSNN is used without Population encoding techniques. In the figure, the calculated SoC is shown as a solid black line, while the solid coloured lines represent the predicted SoC using different encoding techniques.

3.3. Comparison Comparative Analysis of RSNN and Feedforward Neural Network (FNN) Performance

Further, the performance of the RSNN model can be improved by changing the activation function of the hidden layer in the fully connected network to ReLU.

With the same network configuration and a time step of 30 ms, the network achieved a lower average loss of 0.0005 compared to 0.001 when using the sigmoid activation function. Figure 6 shows the performance of the RSNN with ReLU (solid red line) compared to sigmoid (solid blue line). Although the performance is

approximately identical for most of the discharge cycle, the ReLU function enhances the RSNN's accuracy in predicting SoC at both the early discharge stage (near 100% charge) and the later discharge stage (near 0% charge).

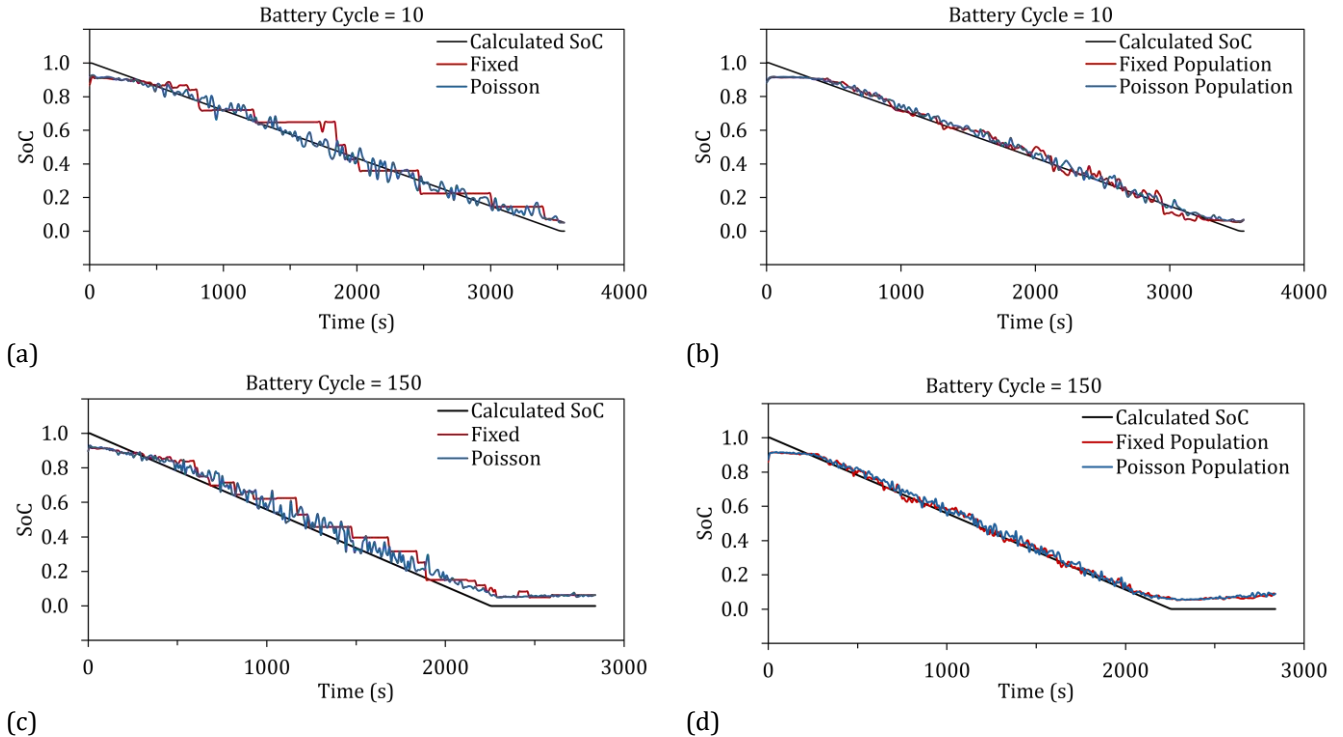


Figure 5. Comparison of SoC using different encoding techniques with a time step of 50 ms: (a) and (b) at the 10th cycle of battery life, and (c) and (d) at the 150th cycle of battery life

Next, a feedforward multilayer perceptron (MLP) was implemented as a baseline model for comparison with the RSNN. The network consists of three fully connected hidden layers, each with 32 neurons using ReLU activation, followed by a dropout layer with a 25% drop rate to reduce overfitting. The input layer receives feature vectors with dimensionality corresponding to the training dataset, while the output layer is a single linear neuron for regression. The model was trained using the Adam optimiser with default parameters and optimised by minimising the mean absolute error (MAE) loss function. This architecture was chosen because ANN are theoretically sufficient to predict SoC, given that the SoC curve decreases approximately linearly over time.

To ensure a fair comparison with the RSNN, both models were trained for the same number of epochs (20) using batch size = 1, since RSNN simulations in BindsNET do not support batch training. In addition, both models were evaluated under reduced training conditions, with the number of training examples randomly subsampled to 50,000, 10,000, 5,000, and 1,000 samples to analyse learning behavior under limited data availability. Generalization performance was assessed using three unseen battery cells. Table 4 summarises the performance metrics of RSNN and FNN. In terms of training time, RSNN requires approximately the same amount of time as FNN. For example, when FNN is trained with a dataset size of 50,000, the training takes 1394 seconds, which is comparable to the RSNN trained with the full dataset. As the dataset size decreases, the

training time of both models is significantly reduced, with RSNN requiring only 24 seconds when trained using 1,000 samples.

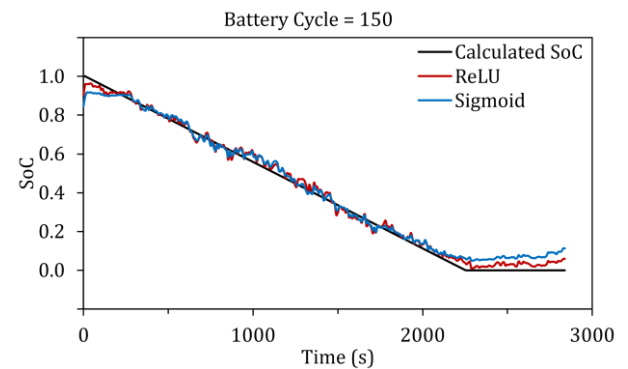


Figure 6. Comparison of SoC prediction between RSNN with ReLU and sigmoid activation function at 150th battery cycle

After 20 epochs, both models achieve comparable prediction accuracy when trained with the full dataset. As the training dataset is reduced, the FNN generally maintains lower RMSE, while the RSNN preserves stable performance and reasonable prediction accuracy even with substantially fewer samples.

As shown in Figure 7, at the 90th (a) and 150th (b) battery cycles, although the FNN produces smoother outputs without fluctuations, it exhibits a large error in SoC prediction during both middle and late discharge

stages. In contrast, the RSNN demonstrates strong temporal tracking capability, capturing the discharge dynamics across different operating phases. These results indicate that although FNN remains competitive in terms of absolute error, RSNN provides a viable alternative framework for SoC estimation, particularly

when temporal representation, data efficiency, and neuromorphic compatibility are considered. Nonetheless, this study focuses only on simulating RSNN using conventional computing with the BindsNET library, which imposes certain limitations and does not fully reflect the potential of RSNN model.

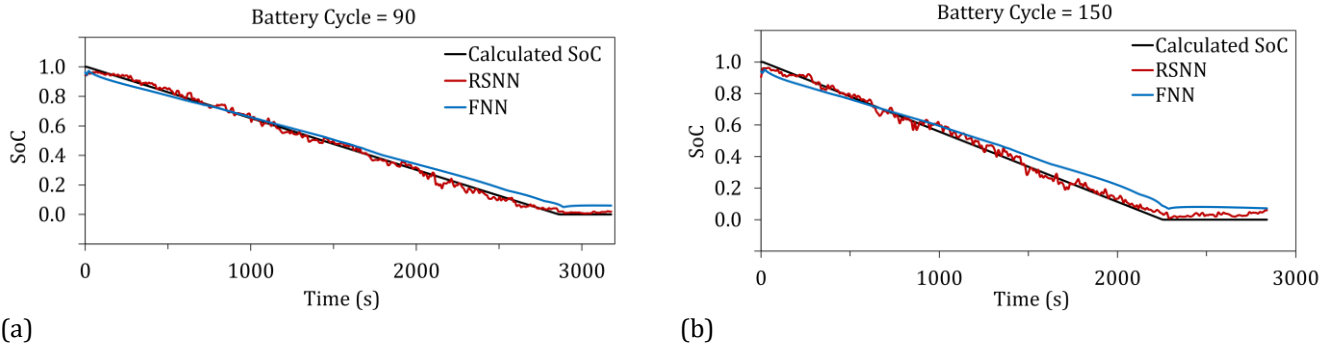


Figure 7. Comparison of SoC prediction between RSNN and FNN model at (a) 90th (b) 150th battery cycle

Table 4. Performance comparison between RSNN and FNN with different number of training data

Dataset Size	FNN (32-32-32)			RSNN (100 nodes)		
	Loss	RMSE	Train Time [s]	Loss	RMSE	Train Time [s]
50000	0.0396	0.037±0.014	1394	0.0004	0.039±0.017	1237
10000	0.0280	0.029±0.006	324	0.0005	0.036±0.012	240
5000	0.0289	0.032±0.011	159	0.0008	0.056±0.023	124
1000	0.0365	0.034±0.004	33	0.0012	0.044±0.001	24

4. Discussion

The findings of this study highlight that the encoding strategy plays a pivotal role in the performance of the RSNN for SoC estimation. The Population-Fixed encoding produced the highest spike diversity and accuracy but at the expense of greater synaptic activity and energy consumption. This result reinforces the inherent trade-off between representational richness and computational efficiency in spiking neural systems. Such a trade-off is particularly critical for real-time, energy-constrained applications such as electric vehicle battery management systems.

In contrast, the Population-Poisson encoding achieved a more favorable balance between accuracy and efficiency, showing that moderate stochasticity in spike generation can improve the network's temporal feature representation while minimising computational load. Such results align with the broader aim of developing neuromorphic models capable of efficient, event-driven computation suitable for embedded or edge platforms. Comparative analysis with the FNN shows that RSNN can achieve prediction performance comparable to conventional neural network model under identical training conditions, while offering explicit temporal processing capability. The improved performance with the ReLU activation function demonstrates the RSNN's capability to maintain predictive stability and generalization while operating under sparse activation regimes. These outcomes suggest that RSNN models are promising candidates for next-generation battery

management systems, combining accuracy with low computational overhead.

Another critical aspect identified through this study is the importance of feature selection and optimization. Although all available input features were used here, not all contribute equally to SoC prediction. Redundant or weakly correlated features may dilute the model's learning efficiency and increase energy consumption. Previous studies, such as the [27], have shown that selecting the most informative features can significantly improve both accuracy and convergence rate. Similarly, in the context of battery systems, works like [28,29] have demonstrated that optimised feature selection not only enhances prediction accuracy but also reduces computational redundancy by focusing on features that strongly correlate with battery degradation or electrochemical behavior. These findings collectively emphasize that incorporating feature selection mechanisms within the RSNN framework could further enhance its performance and adaptability to real-world datasets.

In addition, generalization across different battery chemistries and operating conditions is a critical requirement for practical SoC estimation. Although this study focuses on a single lithium-ion battery dataset, the recurrent spiking reservoir is inherently capable of capturing nonlinear and time-dependent dynamics arising from variations in chemistry, temperature, load profiles, and aging effects. Unlike static feedforward models, the reservoir projects inputs into a rich temporal feature space, which is known to support robust and discriminative representations of nonlinear temporal

dynamics under varying operating regimes [30]. Nevertheless, explicit cross-chemistry and cross-condition validation is beyond the scope of this work and will be pursued in future studies.

Overall, the results indicate that effective SoC estimation using RSNN depends on balancing three key factors: spike diversity, energy efficiency, and input feature relevance. Future work should focus on integrating adaptive encoding and intelligent feature selection techniques to enable dynamic feature prioritization and real-time performance optimization on neuromorphic hardware such as Intel Loihi. Such advancements would accelerate the development of compact, energy-efficient, and intelligent battery management systems for electric vehicles and renewable energy storage technologies.

5. Conclusion

This study successfully achieved its objectives by preprocessing the NASA lithium-ion battery dataset for spiking input and developing a reservoir-based RSNN model. The results showed that different encoding techniques yield distinct spiking dynamics in the reservoir layer. Population-based encoding was essential for accurate SoC prediction as it produced greater diversity in spike count patterns, reflected by higher entropy values. However, this accuracy improvement came at the cost of higher energy consumption measured through increased synaptic events. In addition, applying the ReLU activation function in the hidden layer of the fully connected network significantly improved prediction accuracy, particularly during the middle and late stages of battery discharge where SoC estimation is most critical. Despite some fluctuations in the spiking-based prediction curves, the RSNN captured the nonlinear discharge dynamics more effectively than the FNN with an improvement of approximately 63%. Training efficiency was also found to be comparable between RSNN and FNN, although the scalability was restricted by the limitations of the BindsNET framework which include the lack of batch training and reliance on CPU simulation. Further evaluation of RSNN performance on more complex and noisy datasets with full charge–discharge cycles as well as deployment on neuromorphic systems is therefore essential.

Acknowledgement

The author would like to thank Centre for Research and Innovation Management (CRIM), Universiti Teknikal Malaysia Melaka (UTeM) for its financial support Grant No. PJP/2024/FTKEK/PERINTIS/SA0015 for completing this project.

Author contributions

Muhammad Raihaan Kamarudin: Conceptualization, Methodology, Software, Formal analysis, Investigation, Project administration, Writing-Original draft preparation **Mohd Syafiq Mispan:** Data curation, Validation **Muhammad Noorazlan Shah Zainudin:** Visualization, Investigation, Writing-Reviewing and

Editing **Hazrina Sofian:** Visualization, Investigation, Writing-Reviewing and Editing.

Conflicts of interest

The authors declare no conflicts of interest.

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